HOLIDAY PROGRAMME FOR SENIOR 5 AND SENIOR 6 STUDENTS

Machine Learning Track

Eight weeks - twenty-four sessions - seventy-two hours

What your child will be able to do on the last day

Open a data file they have never seen, clean it, find something true in it, teach a computer to make a prediction from it, check honestly whether the prediction is any good, and stand up for five minutes and explain the whole thing to a room.

They will have built that themselves, on their own laptop, from a starting point of never having written a line of code. It takes seventy-two hours and it is genuinely within reach of an S.5 student.

The work is published to a portfolio they keep, and the certificate carries a mark earned from that work rather than from attendance.

What exists at the end of each week

72 hours - 24 sessions - 3 hours each



The eight weeks, and what exists on their laptop at the end of each one.

1 · The roadmap

Fifteen stages, in the order a person actually learns them. The colour of each card says which fortnight of the track covers it. It is the first thing in this syllabus because a student deserves to see the whole route before they start, including where it stops.



Fifteen stages. The colour says which fortnight covers it. Stage 2 is the mathematics they already have from A-level. Stage 15 is what comes after this track, and it is grey because this track does not teach it.

Two of the fifteen cards are not taught here, and both are deliberate.

Stage 2, the mathematics, is marked as something they already have. The mean, the median, spread, correlation, the line of best fit, probability and gradients are all on the A-level syllabus. This is the most encouraging fact in the whole document: a student who is doing Subsidiary or Principal Mathematics is already holding half of what machine learning is built on, and has never been shown that it is the same thing.

Stage 15 is what comes next. Image recognition, language models and reinforcement learning all need the foundations this track builds. Section 10 says where to learn them.

2 · Week by week

Twenty-four sessions of three hours. Each week below says what is taught, which A-level mathematics it leans on, and what exists at the end of the week. That last column is the one to read if you are short of time.

Week 1 · Your first programs

Roadmap stages 1, 3

#	Session	What happens
1	What machine learning actually is, and your first program	Three live demonstrations: a model that reads handwriting, one that separates spam from real messages, one that guesses a price. Then how each of them is only arithmetic on a table of numbers. Everyone fills in the twelve-question class survey, which becomes the dataset used in Weeks 2, 3, 4 and 7. Set up Python on your own laptop, and write and run a program before you leave.
2	Variables, numbers and text	Storing a value under a name and using it again. The four types you need: whole numbers, decimals, text and true/false. Reading input from the keyboard and printing an answer. Your first error message, on purpose, and how to read it.
3	Making decisions: if, else and comparison	A program that behaves differently depending on its input. Comparison and the words and, or and not. By the end you have written a program that grades a mark out of 100 into a letter.

The maths behind it	Nothing new. You are using arithmetic you have had since P.7.
What you have at the end	Four small programs that run, and Python working on your own laptop.
Before the next session	Rewrite the grading program from memory, without notes. Fifteen minutes, and it is worth more than an hour of reading.

Week 2 · Lists, loops and functions

Roadmap stage 3

#	Session	What happens
1	Lists: holding many values at once	One name holding thirty marks. Getting one item out, adding, removing, counting, sorting. Counting from zero, which is the one real surprise in this week.
2	Loops: doing the same thing to every item	Repeating an instruction for every value in a list without writing it thirty times. Adding a list up, finding the largest, counting how many pass.

#	Session	What happens
3	Functions, and your first real program	Giving a name to a piece of work so you can use it again. Inputs, a result, and why this is the same idea as a formula in Excel. Then you build a marks program: it takes a class list, computes the mean and the highest, and says who passed.

The maths behind it	Mean and highest value. You are writing the formulas you already use in Excel, in a language that can also do the rest of this course.
What you have at the end	A working marks program, and your code saved online where you can show it.
Before the next session	Add the median to your marks program. You know what a median is; the work is getting the computer to find it.

Week 3 · Real data, real files

Roadmap stage 4

#	Session	What happens
1	From spreadsheet to pandas	Opening a real data file in Python, starting with your own class survey so the rows are people you can ask. Rows, columns and the words the field uses for them. Six commands that tell you what any new file contains, in the same order every time; you will use those six on every dataset for the rest of the track.
2	Choosing rows and columns	Filtering to the rows you care about and picking the columns you need. This is the autofilter you already use, written down so it runs again tomorrow on a bigger file.
3	Grouping and summarising	One number per group: the average mark per subject, the total sales per month. This is a pivot table, written down.

The maths behind it	Mean, median, mode and range, applied to a file with thousands of rows rather than the twenty in a textbook exercise.
What you have at the end	A notebook that opens a real dataset and answers five questions about it.
Before the next session	Find a dataset that interests you, from the list we give you, and write down three questions you would like it to answer.

Week 4 · Cleaning, and drawing what you find

Roadmap stages 5, 6

#	Session	What happens
1	Data is never clean when it arrives	Missing values, the same thing spelled three ways, numbers stored as text, and rows recorded twice. Each one found in two lines and fixed in two more. Writing down what you changed, so somebody else can get your number back.
2	Charts that answer a question	Four charts and the question each one answers: a line for change over time, a bar to compare groups, a histogram for the shape of one column, a scatter for whether two things move together. Titles, axis labels and units.
3	Checkpoint 1: your first data investigation	Take a dataset you have not seen, clean it, draw three charts and write five findings with a number attached to each. Marked, and it counts towards your certificate.

The maths behind it	Correlation and the scatter diagram, straight out of the statistics you are taught for A-level. You will draw one and then compute the correlation coefficient.
What you have at the end	A cleaned dataset, three labelled charts and five written findings. The first thing in your portfolio.
Before the next session	Finish and submit the investigation.

Week 5 · Teaching a computer to predict

Roadmap stages 7, 8, 9

#	Session	What happens
1	What learning from data means	Features and a target. Showing a computer examples and letting it find the pattern instead of writing the rule yourself. Splitting your data into a part to learn from and a part to be tested on, and why that split is the whole game.
2	Predicting a number: the line of best fit, by machine	You have drawn a line of best fit by eye. Today the computer finds the one that fits best, and you see that y = mx + c is exactly what a first machine learning model is. Then you use it to predict a value it has never seen.
3	Predicting a category: nearest neighbours	Deciding which group something belongs to by looking at the examples it most resembles. No equations. Then the same idea made to fail on purpose, so you see what it cannot do.

The maths behind it	y = mx + c, gradient and intercept, and the line of best fit. This is the week where A-level algebra turns into a working model.
What you have at the end	Two trained models on your laptop, one predicting a number and one predicting a category.
Before the next session	Retrain your line-of-best-fit model on a different pair of columns and write one sentence on whether it got better or worse.

Week 6 · Is the model any good?

Roadmap stages 9, 10

#	Session	What happens
1	Decision trees, which you can read	A model that makes a decision by asking a series of yes-or-no questions, drawn as a diagram you can follow with your finger. Then you print the tree your data produced and explain it out loud.
2	Accuracy lies: how to actually check a model	Ninety-nine per cent accurate and completely useless. The confusion matrix, and what it costs to be wrong in each direction. Precision and recall, in plain words, on an example where the difference matters.
3	Checkpoint 2: build, test and defend a model	Train two different models on the same data, compare them honestly, choose one and write down why. Marked, and it counts towards your certificate.

The maths behind it	Probability, percentages and rates: the difference between a count and a proportion, which is where most wrong claims about models come from.

What you have at the end	Two compared models, a confusion matrix you can read, and a written justification for the one you chose.
Before the next session	Finish and submit the checkpoint.

Week 7 · Grouping, neural networks, and the real world

Roadmap stages 11, 12, 13

#	Session	What happens
1	Finding groups nobody labelled	Letting a computer sort things into groups when you have not told it what the groups are. Run first on the class survey from Week 1, where you can check the answer against people you know, and then on the penguin measurements, where the groups turn out to be the three species. Same six lines of code, two very different answers.
2	What a neural network is, and where ChatGPT fits	One artificial neuron, worked by hand on paper, then the same thing with thirty of them in three lines of code. What deep learning is, what a large language model is doing when it answers you, and what it is not doing. The honest limits.
3	Machine learning in the world, and choosing your project	Where these methods are already deciding things about people: loans, admissions, policing, medical triage. What bias in a model means and how it gets there. Then you choose your project dataset and question, and write the plan.

The maths behind it	Weighted sums and the gradient of a curve. For Principal Maths students, the derivative you learned last term is exactly how a neural network improves itself.
What you have at the end	A clustering notebook, a small neural network that runs, and a one-page project plan.
Before the next session	Collect or download your project dataset and profile it with the six commands from Week 3.

Week 8 · Your project, and showing it

Roadmap stage 14

#	Session	What happens
1	Build, with help at your elbow	Three hours on your own project. The instructor moves between students; anything asked twice is answered to the whole room. By the end of the session your model trains and produces a number.
2	Finish it, write it up, publish it	The last fixes, then the write-up: your question, your answer with its number, how you know, and what would change your answer. Push the whole thing to your public portfolio.
3	Showcase, and certificates	Five minutes each, in front of the room, with parents invited. The question, the answer, one chart, and what you would do next. Then certificates.

The maths behind it	Everything above, used at once, on data you chose.
What you have at the end	A finished project: a working model, a written page, a public portfolio link and a five-minute presentation.
Before the next session	None. Show somebody at home what you built.

3 · A session, hour by hour

Every session has the same shape, so nobody has to wonder what is coming. These are teenagers on holiday, so no teaching block runs longer than twenty-five minutes and more than half of the time is spent at the keyboard rather than listening.

Minutes	What	Why it is built this way
15	Warm-up	A short puzzle on the screen as they arrive, and last session's homework answered. Starts on time even if the room does not.
25	Teach one idea	One idea, typed live on the projector while they follow on their own machines. Twenty-five minutes is the ceiling.
35	Do it yourself	They build the same thing without being led, working in pairs. The instructor circulates rather than presents.
15	Break	Off screens. Eat something.
25	Teach the second idea	The half that only makes sense once the first half is in their fingers.
45	Build something that runs	Every session ends with a file that does something. Nothing is left half-finished on the screen.
20	Show and tell	Two or three students put their screen on the projector. This is the part they come back for.

Every session ends with something that runs

A student never leaves with a half-finished screen. The last forty five minutes of each session exists so that there is a file on their laptop that does something, however small, before they go home.

The twenty minutes after that is show and tell: two or three students put their own screen on the projector. It is the part they come back for.

4 · The data you will actually work on

A course like this is only as concrete as its data, so here is all of it. Nothing in the list below needs the room's internet to be working, which is deliberate.

Where it comes from	What it is
Your own class, from Week 1	Twelve questions everybody answers in the first session: hours of sleep, distance to school, subject combination, screen time, how you travel, and so on. About twenty-five rows, and every student is one of them, which is why this is the dataset the room argues about. Used in Weeks 2, 3, 4 and 7. No names on it, and it is never published to anybody's portfolio.
Data that arrives inside the software	Handwritten digits, house prices, penguin body measurements and a few others are installed along with scikit-learn, so one line of code loads them and nothing is downloaded. These are the standard teaching datasets used in universities, which is deliberate: a student who meets them here will meet them again next year.
Ugandan open data, on the USB stick	Health facility locations, population and school enrolment by district, and a rainfall series. All of it is published free by public bodies and humanitarian data services, downloaded before the track starts and handed out on day one. Working on data about places students know is worth more than a tidier dataset about somewhere they do not.
Data the student collects for the project	Boda fares against distance, second-hand phone prices against their specifications, match results, a market's prices over three weeks. Forty rows gathered by hand in a notebook is plenty for Week 8, and a student defends their own data far better than anything handed to them.
Kaggle, for after the track	Thousands of free public datasets and beginner competitions. Week 8 shows how to enter one, and it is where a student who wants to keep going finds their next dataset.

The class survey stays in the class

The dataset the students spend the most time on is the one they make in the first session, because a table with your own sleep and your own journey to school in it is the one you argue with. It carries no names, it is not published to anybody's portfolio, and it is deleted at the end of the track.

Everything students do publish is built on public data, or on data they collected themselves.

5 · Project ideas, so you can picture yours

Students choose their own project in Week 7 and present it in Week 8. These are examples to start an argument with rather than a menu, and the best projects every time are the ones a student brought themselves. Each one names where the data would come from, because an idea with no dataset behind it is a wish.

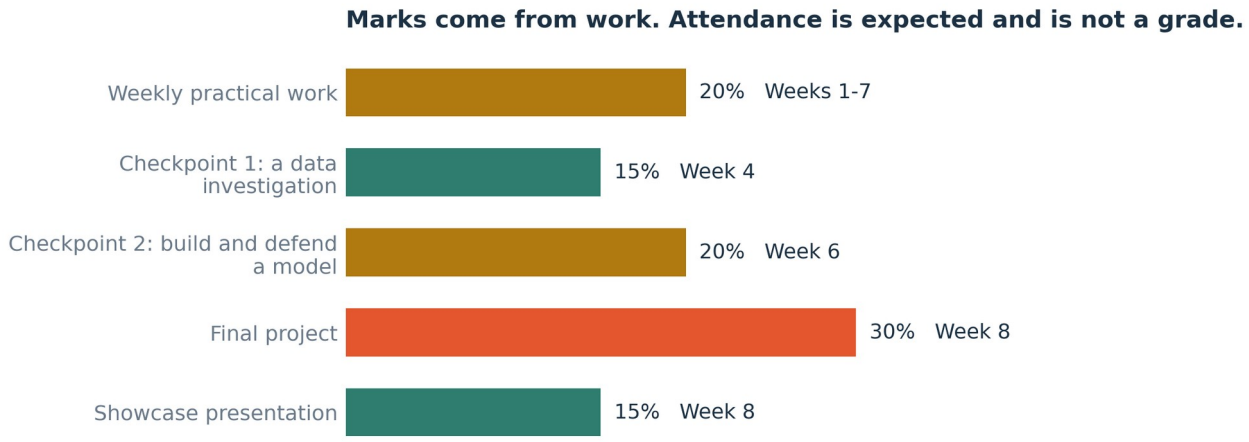
The question	Where the data comes from
Does more sleep go with better marks?	The class survey, plus your own marks for the last three terms.
What is a second-hand phone worth, given its age, memory and screen?	Forty listings you copy from an online marketplace into a spreadsheet.
Can you predict a boda fare from the distance and the time of day?	Fares you record yourself for two weeks, with the distance from a maps app.
Which subject combinations go together in your school, and why?	Subject choices for your year, with no names attached.
Can a computer read a handwritten digit?	The handwritten digit images that arrive with scikit-learn.
Does rainfall predict the price of tomatoes in your market?	The rainfall series on the USB stick, and prices you record weekly.
Which districts are furthest from a health facility?	The health facility and population data on the USB stick.
Can you tell a real message from a scam message?	A public collection of scam and ordinary text messages, on the stick.
Does a footballer's minutes played predict their goals?	A season of public match data for a league you follow.

What a project has to satisfy

1	Your question must have a number or a category as its answer. “Is machine learning useful in farming” is an essay; “can I predict maize yield from rainfall” is a project.
2	The data must exist before you commit to the question. You profile it in Week 7 with the six commands from Week 3, in front of the instructor.
3	About two hundred rows if you download it, or forty if you collect it yourself. Below that, nothing you find will hold up.
4	No personal information about anybody who has not agreed to be in it, and no names in anything you publish.
5	You must be able to explain every line you submit, including any line an AI tool wrote for you.

6 · Marks, and what the certificate means

A certificate with real marks was asked for, so the marks come from work rather than from turning up. Attendance is expected and is not a grade.



Where the hundred marks come from, and when each piece is due.

What	Weight	When	How it is marked
Weekly practical work	20%	Weeks 1-7	Seven short pieces, one a week, marked out of 10. These are the programs and notebooks built in the sessions themselves, so a student who attends and works collects most of this.
Checkpoint 1: a data investigation	15%	Week 4	Clean an unfamiliar dataset, draw three charts and write five findings with a number attached to each. Marked on the findings and the labels on the charts.
Checkpoint 2: build and defend a model	20%	Week 6	Train two models on the same data, compare them honestly and justify the one you chose. Marked on the reasoning as much as on the accuracy.
Final project	30%	Week 8	A working model on a dataset the student chose, with a written page: the question, the answer with its number, how they know, and what would change the answer.
Showcase presentation	15%	Week 8	Five minutes in front of the room with parents present. Marked on whether somebody outside the room could follow it.

What the certificate says

Band	Mark	What it claims the student can do
Distinction	70% and above	Can start a new dataset unaided and finish it.
Merit	60% and above	Can do the whole workflow with the notes open.
Pass	50% and above	Can follow the workflow and explain what each step does.
Certificate of attendance	below 50%	Attended and took part. The certificate says exactly that and claims no skill beyond it.

Late work, missed checkpoints, and second attempts

When you get a mark back	Within a week of handing it in, with written comments on the work itself rather than a number on its own. The two checkpoints are returned in the next session, because a mark that arrives after the topic has moved on teaches nothing.
If you hand something in late	Practical work is accepted up to a week late at full marks, because this is a holiday and life happens. After that it is marked and capped at a pass. The final project cannot be late, because the showcase is a fixed day and the room is booked.
If you miss a checkpoint entirely	You sit it in the following session from a different dataset. A student who misses both checkpoints and the project has no marked work, and the certificate then records attendance only.
If you want to improve a mark	Practical work can be redone once, any time before Week 8, and the better mark stands. The checkpoints and the project are marked once. Redoing work you got wrong is the most useful thing a student on this track can do with an evening.

A student below 50 still receives a certificate, and it does not overstate

It states that they attended and took part, and it does not claim a skill that is not there. We would rather a certificate from this track meant something to the person reading it, including to an admissions officer who has seen a hundred of them.

7 · What to bring

Students bring their own laptops, which means a mixture of Windows and Mac of different ages. Everything below is planned around that.

A laptop they bring themselves	Windows 10 or newer, or any Mac from 2017 onward. 4 GB of memory is enough; 8 GB is comfortable. About 6 GB of free disk space. A student without a laptop should tell us before Week 1 so we can pair them and arrange lab access, rather than discovering it on the first morning.
Nothing installed beforehand, and this matters	We install everything together in Session 1 from a USB stick, so a slow connection cannot stop the first morning. Please do not install Python at home first: the whole class runs one fixed version, and a laptop carrying a different one produces errors nobody else in the room can reproduce and that cannot be fixed during a session. Every piece of software we use is named in the next section.
Two free accounts, made in Session 1	A Google account, for the browser fallback, and a GitHub account, for the portfolio. Both are free and both are set up in the room. A student under thirteen needs a parent to make the GitHub one.
A notebook and a pen	Several things in this track are worked on paper first, including the neuron in Week 7. That is deliberate.
The data, on a USB stick	Every dataset is handed out on the first day as well as being downloadable. The internet in the room is not a dependency for any session.

No laptop? Tell us before Week 1, not on the first morning

A student without a laptop can still take the track: they pair with somebody for the session work and use a lab machine for the homework. That has to be arranged in advance, and a student who says nothing and arrives with no machine loses the first session.

Everything also runs in a browser with nothing installed, which is the fallback for a laptop too old or too locked down for the install.

8 · The software, every piece of it by name

All of it is free. None of it is a trial, none of it expires, and nothing here asks for a card. It is also the same stack a first-year university course and a working analyst use, so none of it is a toy version that has to be unlearned later.

Tool	What it is for
Python	The language. One version is fixed for the whole class and every laptop gets that one, so an error on your screen is an error the person next to you can also read.
Anaconda	The installer that puts Python and everything below on a laptop in one go, with nothing left to configure. It runs from the USB stick we bring, so the room's internet cannot stop the first session. About 5 GB once installed.
Jupyter Notebook	Where all the work happens: you type a few lines, run them, and see the answer or the chart underneath. A notebook is also the thing you submit, because it shows the working as well as the result.
pandas	Tables in Python. Everything you can do to a spreadsheet, written down so it runs again tomorrow on a file ten times the size.
matplotlib	The charts. Line, bar, histogram and scatter, with the titles and the axis labels that make a chart mean something.
scikit-learn	The models. Regression, nearest neighbours, decision trees, clustering and the small neural network, all in the same three lines of code: fit, predict, score.
Google Colab	The fallback. The same notebooks run in a browser with nothing installed, which covers a laptop that is too old or too locked down for the installer. Needs a free Google account.
GitHub	The portfolio. A free account, and each finished piece of work is pushed there as a public link the student keeps and can put on an application.

One fixed version for the whole class

Please do not install Python at home before Week 1, however keen a student is. The packages this track uses are compiled against a particular version of Python, so a laptop carrying a different one produces an error message nobody else in the room can reproduce, and there is no way to fix it while a session is running.

We install the same version on every machine in Session 1, from a USB stick, and a student who arrives with nothing installed is the easiest person in the room to help.

9 · Questions parents ask

Attendance

Twenty-four sessions, and the work is cumulative: Week 5 assumes Week 2. A student who misses a session gets the notes and the recording of the live coding, and should expect to spend an hour catching up before the next one. Missing more than four sessions makes the certificate marks difficult to award honestly.

Using AI tools such as ChatGPT

Allowed, and taught, with one rule: a student must be able to explain every line they submit. We ask in the showcase and at the checkpoints. Code a student cannot explain scores nothing, which is the same rule any university will apply to them next year. Week 7 covers what these tools are actually doing and where they are wrong.

Safety online

Students create a public portfolio, so the setup session covers what goes in it and what does not: no full date of birth, no home address, no school ID numbers, no photographs of other students without asking. Datasets are public and contain no personal information about anybody in the room.

How much work there is at home

Fifteen to twenty minutes after most sessions, which is one small exercise and is listed in the week-by-week table. Three weeks are heavier because a marked piece is due: Week 4, Week 6 and Week 8 need about two hours each at home. Over the whole track that is roughly ten hours outside the room, and a student who does none of it will still pass and will get noticeably less out of it.

Screens and breaks

Every session has a fifteen-minute break away from screens, and no teaching block runs longer than twenty-five minutes.

What parents are invited to

The showcase on the final session, where each student presents for five minutes. Parents are welcome to the first fifteen minutes of Week 1 Session 1 as well, to see the room and meet the instructor.

Marks, and what the certificate says

Marks come from work rather than attendance. The certificate states the band and the hours, and a student below 50 receives a certificate of attendance that does not claim a skill they do not have. We would rather a certificate meant something.

10 · What this track does not do

Written down because a family deciding how to spend a holiday and a fee deserves the edges as well as the promises.

It does not make your child a professional in eight weeks

Seventy-two hours makes a confident beginner who can finish a small project without help. A working data scientist has years behind them. What this track does is remove the barrier: after it, the online courses and university modules are readable rather than baffling.

It is not a substitute for A-level revision

It is a holiday track and it is scheduled three days a week for that reason. The maths it uses is A-level maths, so it reinforces statistics and algebra, and several students find their Paper 2 statistics easier afterwards. It does not teach the syllabus.

There is no deep learning beyond a first look

Image recognition, language models and reinforcement learning need foundations this track is building. Week 7 shows what they are and Stage 15 of the roadmap says where to learn them. A track that promised them in eight weeks from zero would be selling something.

Nobody is promised a job or a scholarship

Students leave with a public portfolio and a certificate with marks on it, which is a real thing to show an admissions officer. It is an advantage rather than a guarantee, and we will not describe it as more than that.

Where to go after this track

If they want more machine learning	Stage 15 of the roadmap: computer vision, language models, reinforcement learning. Kaggle Learn and Google's machine learning crash course both cover the next step for nothing, and both assume exactly what this track leaves a student with.
If they are heading for university computing	This track covers roughly the practical half of a first-year introduction to programming and data. Keep the portfolio and keep adding to it.
If they are heading for engineering, medicine or economics	Every one of those fields now uses these methods on its own data. The data handling in Weeks 3 and 4 is the part they will use first, whatever they study.
If they want to keep practising for free	Kaggle has thousands of public datasets and beginner competitions. The final week shows them how to enter one.

11 · Practical details

The teaching content of this syllabus is settled. The arrangements below are for the school to confirm, and they are left blank rather than guessed, because a syllabus that goes home with a date in it that turns out to be wrong is worse than one with a gap.

To confirm	Details
Start date and end date	
Which three days of the week, and the time of day	
Venue and room	
Fee, what it includes, and how to pay	
Instructor name and one line of background	
A phone number and an email for questions	
Maximum number of students in the group	
Registration deadline and how to register	

One request of parents, and it is the one that matters most

Ask your child to show you what they built, once a week. Not to check it, and you do not need to understand the code. Explaining it to somebody who does not already know is the single thing that turns a holiday course into a skill that stays, and ten minutes at the kitchen table does more for them than any extra session we could add.